

Statistical Inference and Multivariate Analysis (MA324)

LECTURE SLIDES
Topic 05
Generalized Linear Regression

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Generalized Linear Regression

Introduction to GLMs

Ordinary Least Squares (OLS) regression assumes continuous, normally distributed outcomes. **Generalized Linear Models (GLMs)** extend this framework to accommodate non-normal distributions (e.g., binary, counts).

A GLM consists of three core components:

- 1 **Random Component:** The probability distribution of the response variable Y (must belong to the exponential family).
- 2 **Systematic Component:** The linear predictor $\eta = X\beta$.
- 3 **Link Function:** A function $g(\cdot)$ that relates the expected value of Y , $\mu = E(Y)$, to the linear predictor: $g(\mu) = \eta$.

Logistic Regression

When to Use Logistic Regression

Logistic regression is used when the response variable Y is **binary** or dichotomous (e.g., 0/1, Success/Failure, Dead/Alive).

Why OLS fails for binary data:

- **Non-normality:** The error terms cannot be normally distributed; they only take two possible values.
- **Heteroskedasticity:** The variance of a binomial distribution is $p(1 - p)$, which changes as the probability p changes.
- **Out-of-bounds Predictions:** OLS can predict probabilities < 0 or > 1 .

Logistic Regression: Theory

- **Random Component:** $Y \sim \text{Bernoulli}(p)$, where $p = P(Y = 1)$.
- **Link Function:** The **logit** function (log-odds).

$$\text{logit}(p) = \ln \left(\frac{p}{1-p} \right) = \beta_0 + \beta_1 X_1 + \cdots + \beta_k X_k$$

To solve for the predicted probability p , we use the inverse logit (logistic) function:

$$p = \frac{\exp(X\beta)}{1 + \exp(X\beta)}$$

MLE for Logistic Regression

Instead of OLS, logistic regression parameters are estimated using **Maximum Likelihood Estimation (MLE)**.

For n independent observations where $Y_i \sim \text{Bernoulli}(p_i)$, the probability function is $p_i^{y_i} (1 - p_i)^{1-y_i}$. The **log-likelihood function** (ℓ) is:

$$\ell(\beta) = \sum_{i=1}^n [y_i \ln(p_i) + (1 - y_i) \ln(1 - p_i)]$$

We substitute p_i with our model $\frac{\exp(X_i\beta)}{1+\exp(X_i\beta)}$:

$$\ell(\beta) = \sum_{i=1}^n [y_i(X_i\beta) - \ln(1 + \exp(X_i\beta))]$$

Note: There is no closed-form algebraic solution for β . Algorithms like Iteratively Reweighted Least Squares (IRLS) are used to find the maximum numerically.

Interpreting Coefficients

Unlike OLS, coefficients in logistic regression represent changes in the **log-odds** of the outcome.

- β_1 : A one-unit increase in X_1 is associated with a β_1 increase in the *log-odds* of $Y = 1$.
- e^{β_1} (**Odds Ratio**): A one-unit increase in X_1 multiplies the odds of $Y = 1$ by a factor of e^{β_1} .
 - If $e^{\beta_1} > 1$: The predictor increases the probability of the outcome.
 - If $e^{\beta_1} < 1$: The predictor decreases the probability of the outcome.

Applied Example: Logistic Regression

Using R's built-in `mtcars` dataset, let's predict if a car has a V-shaped engine (`vs = 0`) or a straight engine (`vs = 1`) based on its fuel efficiency (`mpg`).

```
# Fit the logistic regression model
logit_model <- glm(vs ~ mpg, data = mtcars,
                   family = binomial(link = "logit"))

# View coefficients
coef(logit_model)
(Intercept)          mpg
-8.8330546      0.4304135
```

Interpreting the Output

```
# Extract the coefficient for mpg and  
# calculate the Odds Ratio  
exp(coef(logit_model) ["mpg"])  
      mpg  
1.537898
```

Interpretation:

For every 1 mpg increase in fuel efficiency, the log-odds of a car having a straight engine increase by 0.43.

Exponentiating this coefficient gives us an **Odds Ratio of 1.53**. This means that for every 1 unit increase in mpg, the odds of the car having a straight engine increase by 53%.

Poisson Regression

When to Use Poisson Regression

Poisson regression is used when the response variable Y represents **count data** (e.g., number of hospital visits, number of traffic accidents).

Characteristics of count data:

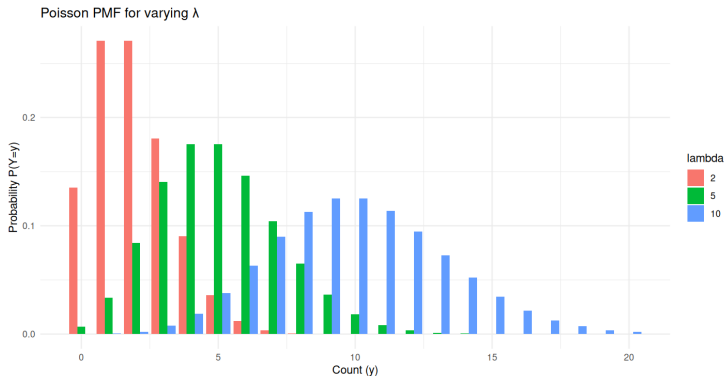
- Discrete, non-negative integers $\{0, 1, 2, \dots\}$.
- Often right-skewed (many low counts, few high counts).

Poisson Regression: Theory

- **Random Component:** $Y \sim \text{Poisson}(\lambda)$, where λ is the expected count (rate).
- The **PMF** is

$$P(Y = k) = \frac{e^{-\lambda} \lambda^k}{k!} \quad \text{for } k = 0, 1, 2, \dots$$

- $E(Y) = \text{Var}(Y) = \lambda$.



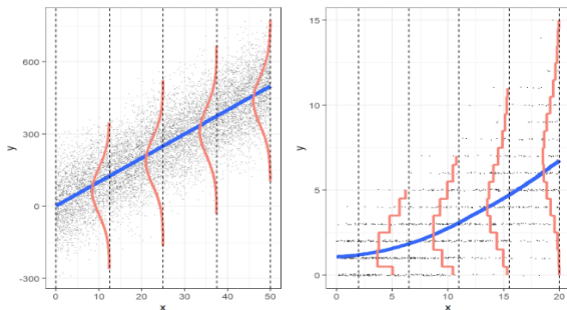
Poisson Regression: Theory

- **Link Function:** The **natural log** function.

$$\ln(\lambda) = \beta_0 + \beta_1 X_1 + \cdots + \beta_k X_k = X\beta$$

- To solve for the expected count λ , we exponentiate the linear predictor:

$$\lambda = \exp(X\beta)$$



Comparison between LR and PR

MLE for Poisson Regression

The **log-likelihood function**:

$$\ell(\beta) = \sum_{i=1}^n [y_i \ln(\lambda_i) - \lambda_i - \ln(y_i!)]$$

We substitute the expected count λ_i with our model $\exp(X_i\beta)$:

$$\ell(\beta) = \sum_{i=1}^n [y_i(X_i\beta) - \exp(X_i\beta) - \ln(y_i!)]$$

Note: As with the logistic model, we must take the derivative with respect to β and use numerical methods (like IRLS) to solve for the maximum.

Interpreting Coefficients

Coefficients in Poisson regression represent changes in the **log-count**.

- β_1 : A one-unit increase in X_1 is associated with a β_1 increase in the *log-expected count*.
- e^{β_1} (**Incidence Rate Ratio / IRR**): A one-unit increase in X_1 multiplies the expected count by a factor of e^{β_1} .

Data Overview: The `warpbreaks` Dataset

To demonstrate Poisson regression, we will use R's built-in `warpbreaks` dataset. It records the number of warp breaks in yarn during weaving under different experimental conditions.

Variables in the dataset:

- `breaks`: (Numeric) The number of breaks per loom. This is our non-negative count response variable (Y).
- `wool`: (Factor) The type of wool used (Levels: A, B).
- `tension`: (Factor) The level of tension applied to the yarn (Levels: L, M, H for Low, Medium, High).

Modeling Goal:

To estimate how changing the `tension` level affects the expected count of `yarn breaks`.

Applied Example: Poisson Regression

Using the `warpbreaks` dataset, let's model the number of warp breaks per loom (`breaks`) based on the level of `tension` (Low, Medium, High).

```
# Fit the Poisson regression model
poisson_model <- glm(breaks ~ tension, data = warpbreaks,
                     family = poisson(link = "log"))

# View coefficients
coef(poisson_model)
(Intercept)      tensionM      tensionH
  3.5945693    -0.3662283    -0.3175402
```

Interpreting the Output

```
# Calculate the Incidence Rate Ratios (IRRs)
exp(coef(poisson_model))
(Intercept)      tensionM      tensionH
  36.3999996    0.6933446    0.7279361
```

Interpretation:

The baseline category for tension is "Low".

The IRR for `tensionM` (Medium) is 0.69. This means that, compared to low tension, changing to medium tension multiplies the expected number of breaks by 0.69 (a 31% decrease in breaks).

Some Extensions

In real-world data, the strict assumptions of these GLMs often fail. Advanced modeling techniques address these violations:

- **Overdispersion:** If $\text{Var}(Y) > E(Y)$, the Poisson assumption is violated, leading to underestimated standard errors.
Solution: Use Negative Binomial Regression or Quasi-Poisson models.
- **Excess Zeros:** If data contains more zeros than the Poisson distribution allows.
Solution: Zero-Inflated Poisson models.

References

References

- <https://bookdown.org/roback/bookdown-BeyondMLR/ch-poissonreg.html>